

AI and ML for Us

Lecture 1:

Introduction to AI & ML

Pradumn K Pandey

CSE, Indian Institute of Technology Roorkee

Today...

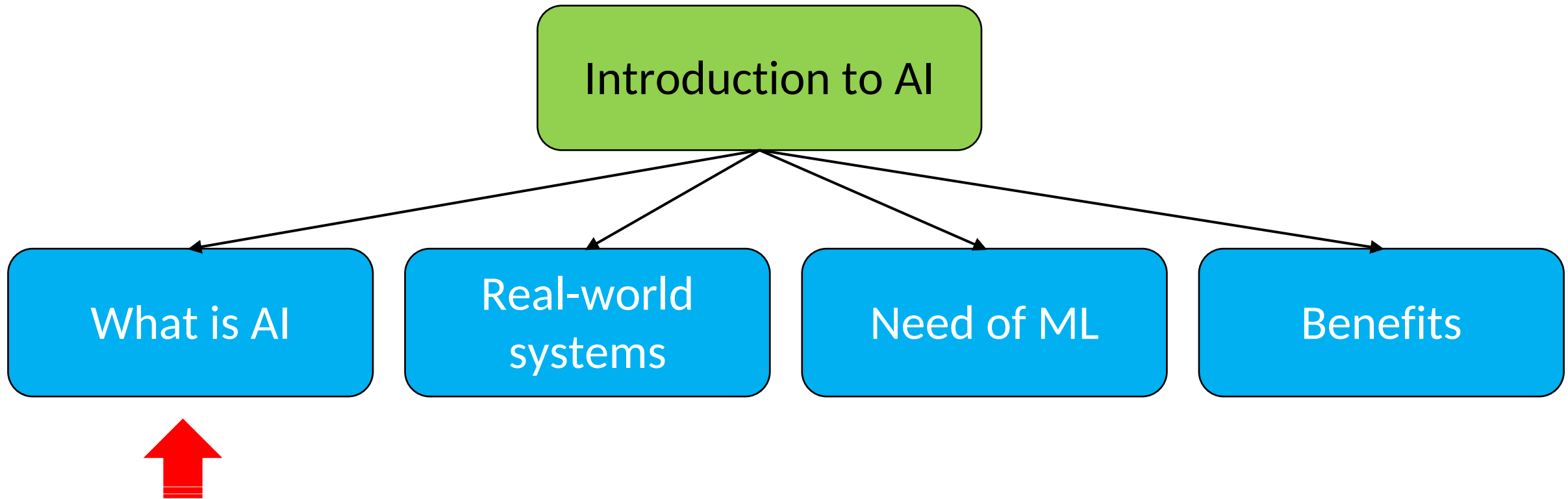
- Today's Session:

- What is Artificial Intelligence (AI)?
- Real Life examples
- Machine Learning
- Benefits of AI

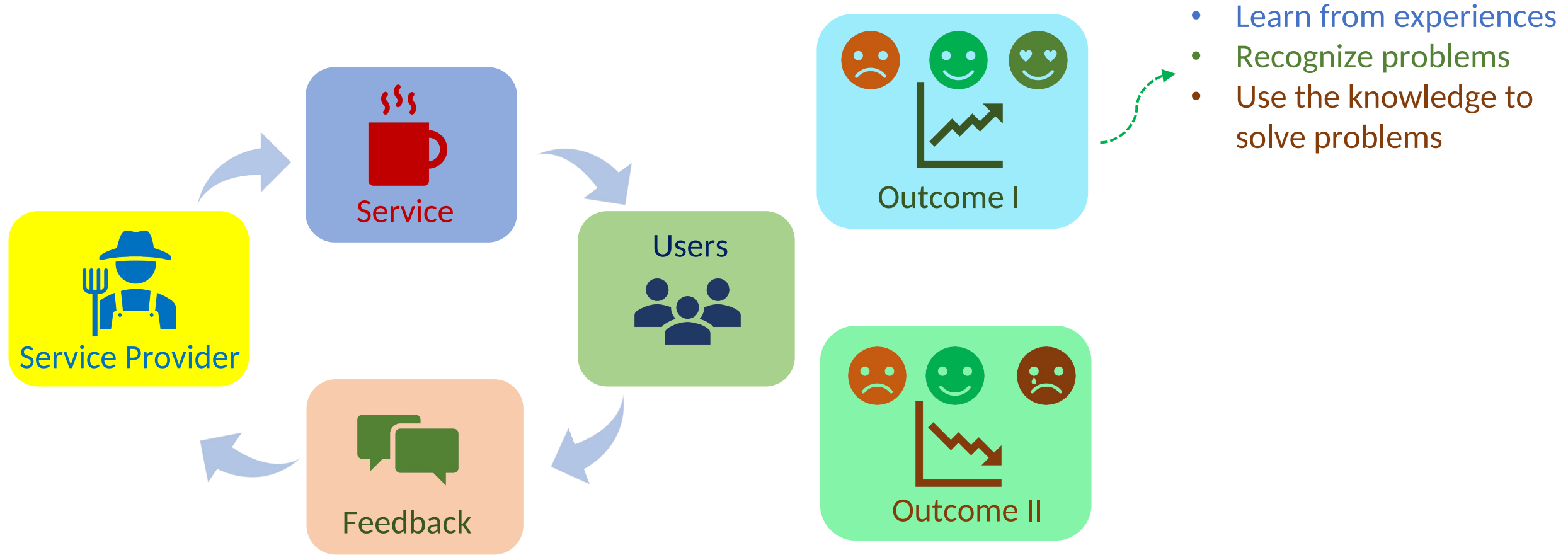
- Announcements:

- Practical Assignments which include preparation of PPT on some given topic (before MTE) and a successful implementation of a given AI project (before ETE)
- Quiz I, II, III before MTE and Quiz IV, V, VI,... after MTE (before ETE)
- 25% Quiz + 25% MTE + 50% ETE

Outline



How to make a better Coffee☕ ?



What is Intelligence?

- **Intelligence** is the
 - Ability to learn from experiences,
 - Adapt to new situations and recognize problems,
 - And use that knowledge to solve problems or make decisions.

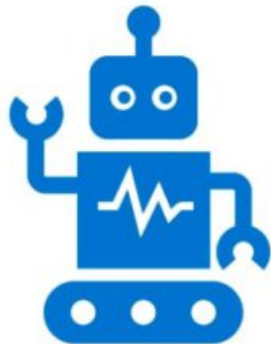


What is AI?

- **Artificial Intelligence (AI)** is the branch of science and technology that is devoted to the creation of machines (Computer System, Robots, etc.) that learn and think as intelligently as human beings is known as Artificial Intelligence or AI.



Automation



Robotics



Virtual Assistants

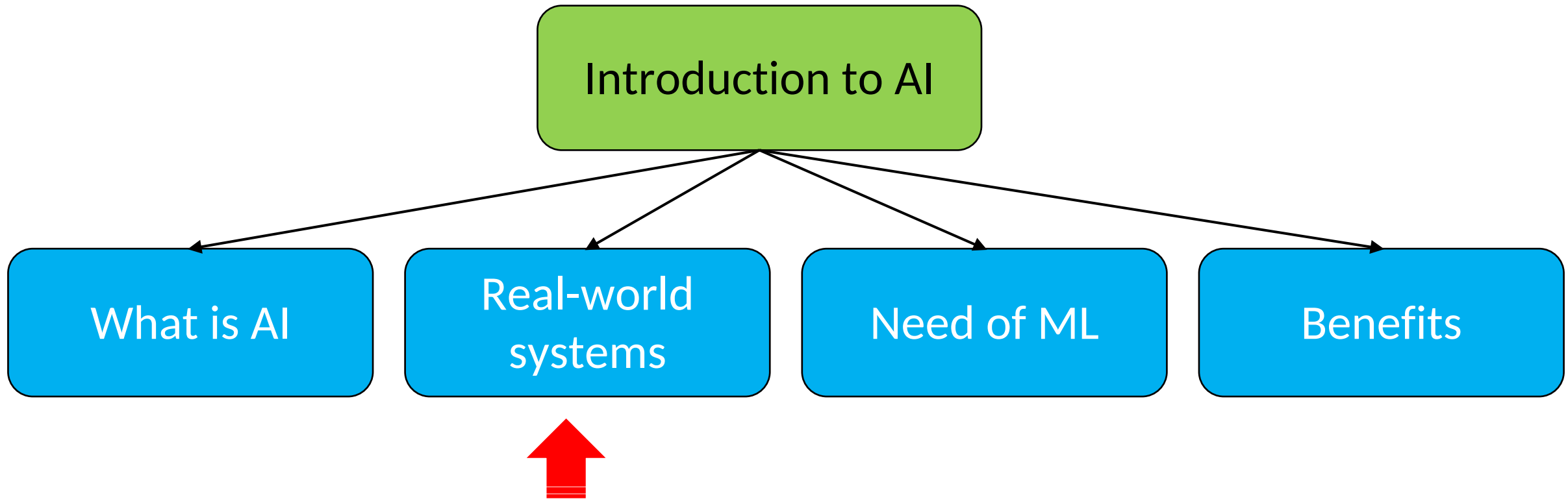


Softwares



Self Driving cars

Outline



E-commerce: Amazon product recommendation



Brand: BOBO BIRD
BOBO BIRD Mens
Personalized Engraved
Wooden Watches, Stylish
Wood & Stainless Steel
Combined Quartz Casual
Wristwatches for Men Family
Friends Customized Gift
4.8 2,677 ratings |
[Search this page](#)

₹727 /month (24 months)
with EMI on your ICICI Bank Credit Card
[All EMI Plans](#) ▼

-30% ₹14,848
M.R.P.: ₹21,219
Inclusive of all taxes
With Amazon Business, you would have saved
₹683.77 in the last year. [Create a free account](#)



Recommended for you



13% OFF
Vim Matic Dishwash
Detergent Powder, 1 kg |
Suitable For All Dishwashers |
With Powerful Enzymes To...
3.9 2,748

Add



Fortune Rinse Aid Liquid-
500 ml
4.4 2,055
₹200⁰⁰

Add



9% OFF
Emami Healthy and Tasty
Refined Rice Bran Oil Jar,
4.5kg/5L
4.4 386
₹773⁰⁰ ₹850.00

Add

Products related to this item

Sponsored



FORSINING
Silicone,Rubber Mens
Winner Gold Skeleton
Triangle Watch Man...
★★★★☆ 4
₹5,399⁰⁰



Fastrack Space Rover
Quartz Analog Black Dial
Silicone Strap Unisex
Watch-NS38003PP25
★★★★★ 1
₹647⁰⁰



Styling ideas



Pepe Jeans
Men's Skinny Jeans
★★★★☆ 10
₹2,708⁰⁰



ADRO
★★★★★ 5
₹999⁰⁰



Puma
★★★★★ 80
₹2,999⁰⁰
[+1 colour/pattern](#)

Entertainment: Movie recommendation

NETFLIX

HomeTV ShowsMoviesNew & PopularMy ListBrowse by Languages

SearchChildren4

Today's Top Picks for You



Recently added



Recently added



Recently added





Recently added



Continue Watching for PRADUMN PANDEY >



New EpisodeWatch Now







TOP 10Recently added



Leaving Soon



US TV Shows Dubbed in Hindi













Virtual Assistant: Amazon Alexa



User

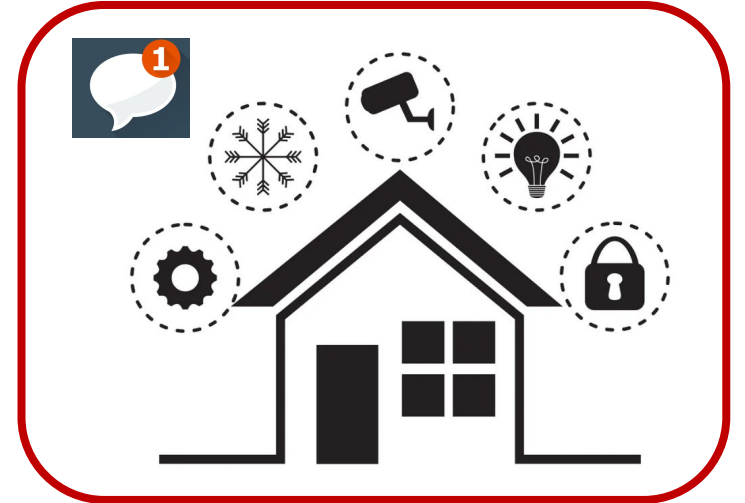
alexa



Music



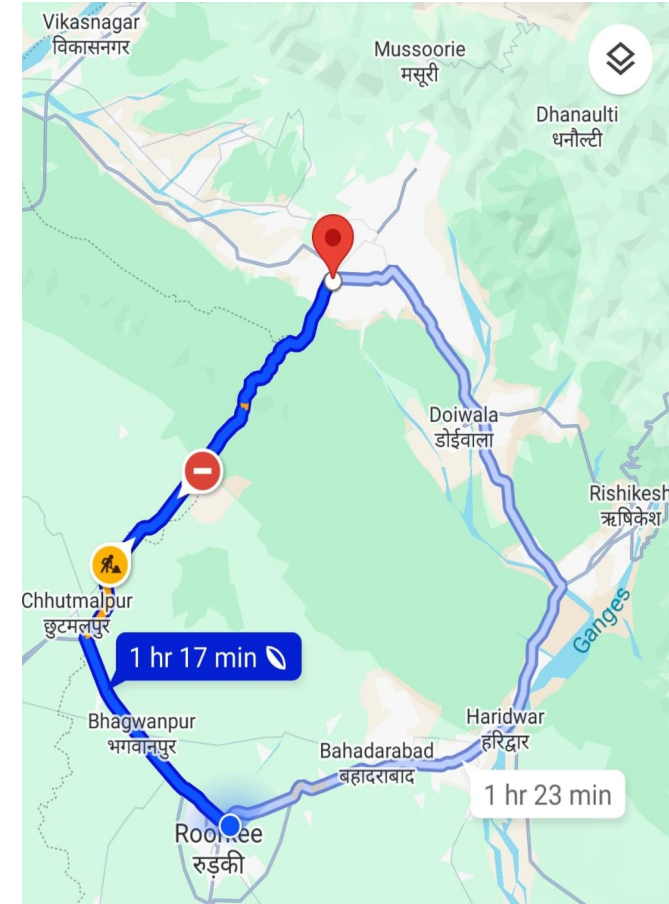
Smart Home



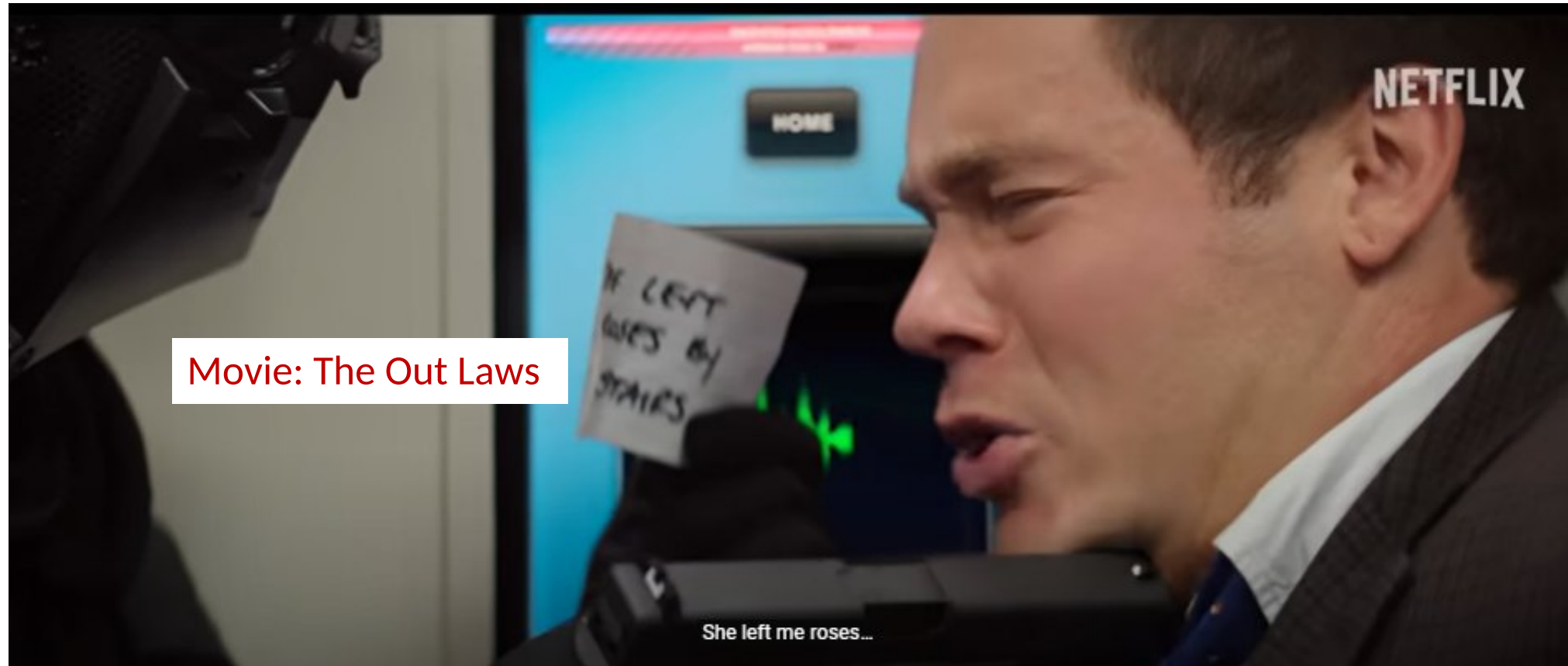
Vehicle



Google Traffic Prediction



Security: Your Device **knows** You 😊



Movie: The Out Laws

- An example of bank vault opening using voice command

Help and Learning: ChatGPT



correct the following text: I don know how I can write a better peper

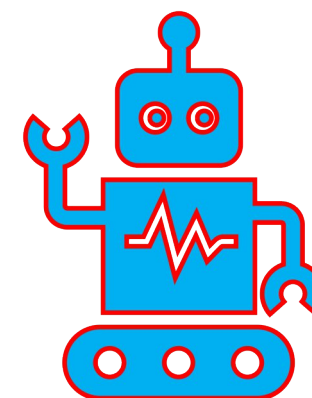
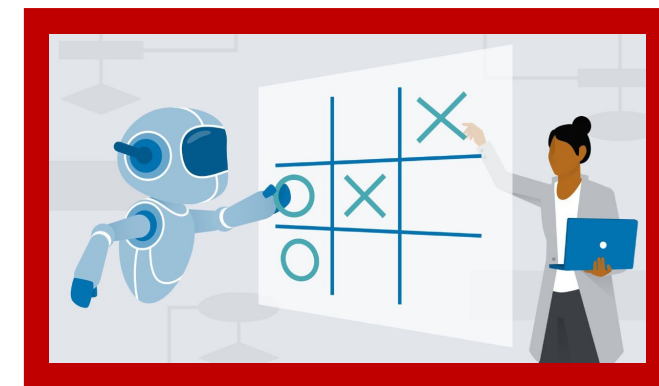
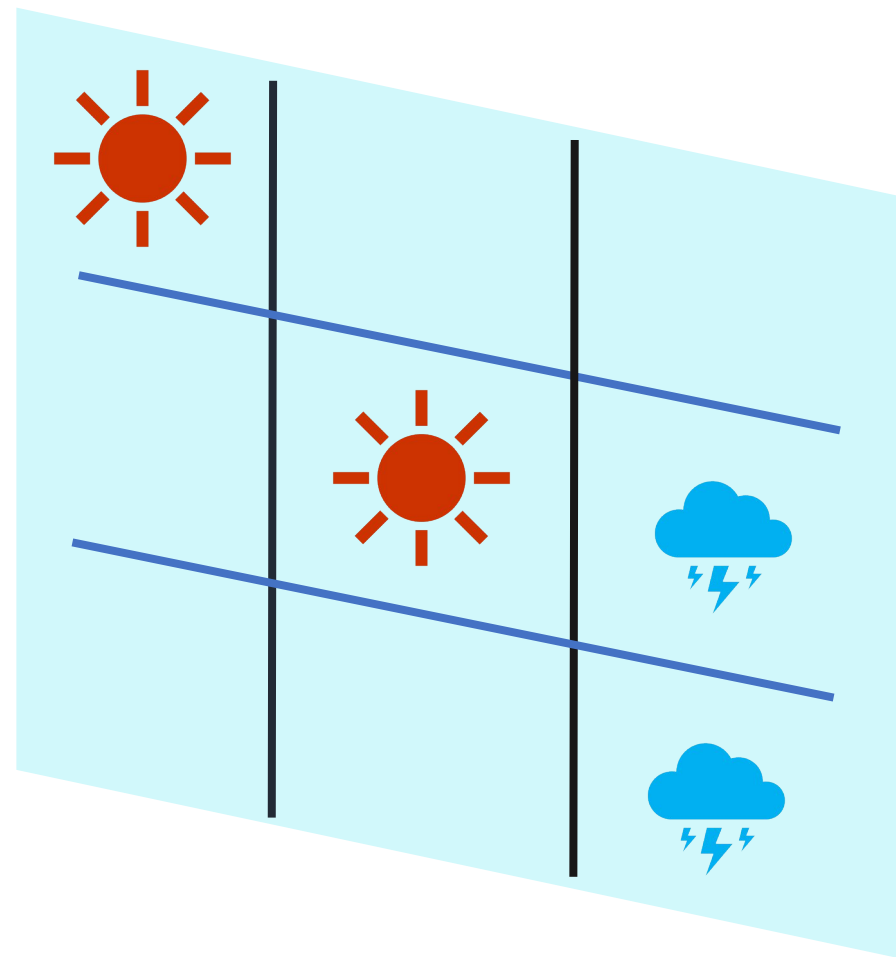


Here's the corrected text:

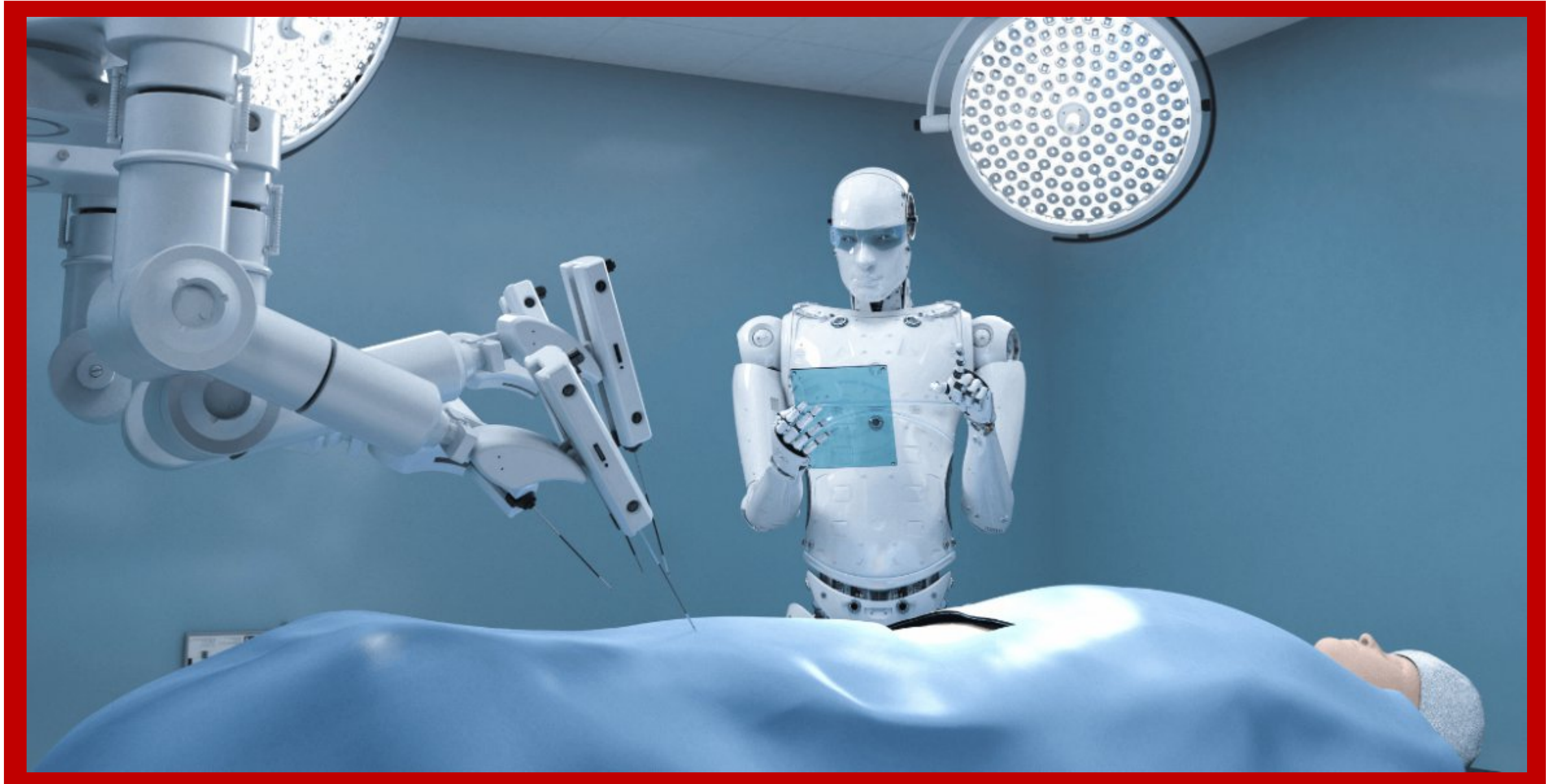
I don't know how I can write a better paper.



Gaming: Tic-Tac-Toe



AI4Healthcare



AI Chatbot



Hello, Chatbot

Hello,
How can I help you?

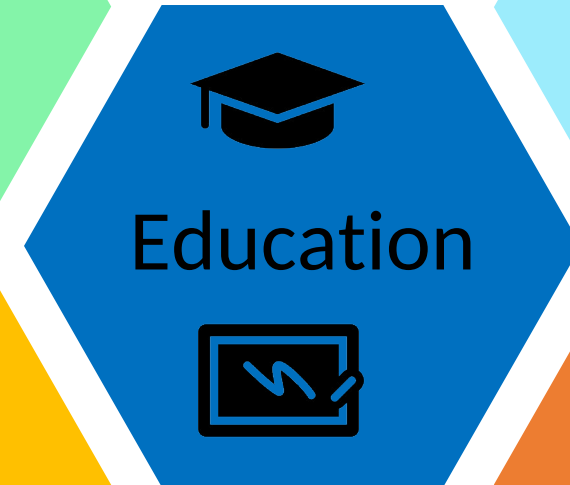


AI in Education

Ways Artificial Intelligence Will Change Education



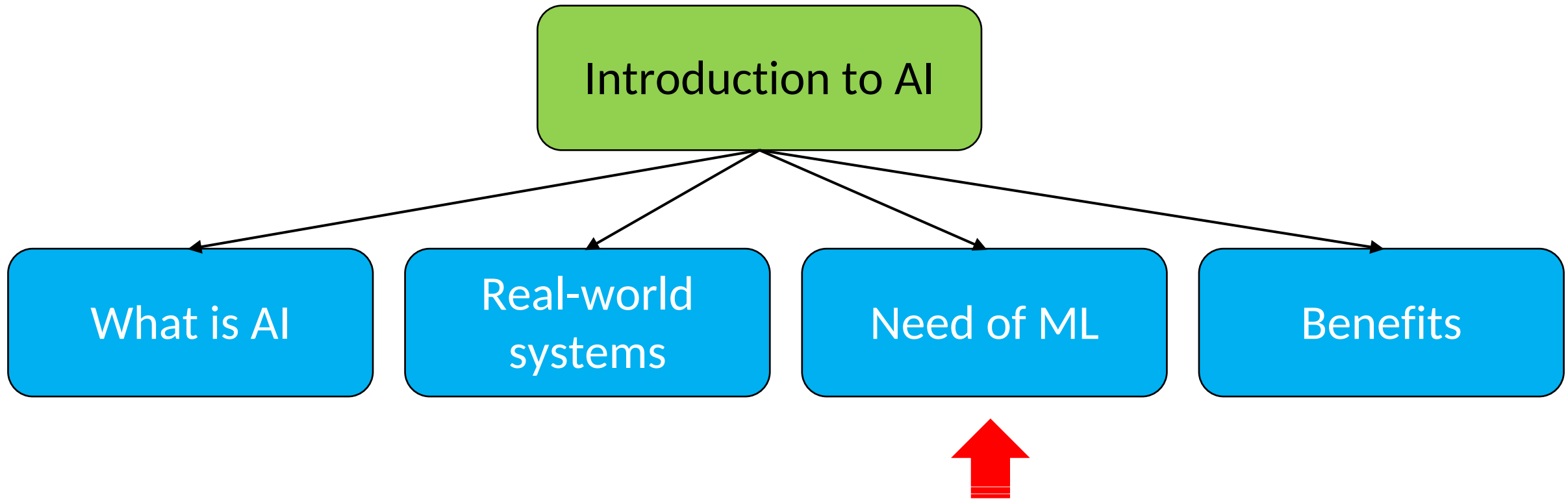
1. Personalized Learning Paths
2. Identifying Learning Gaps
3. Intelligent Feedback & Assessment
4. Creating Engaging & Interactive Content
5. Empowering Educators
6. Making Education More Accessible



AI in Self-Driving Cars



Outline

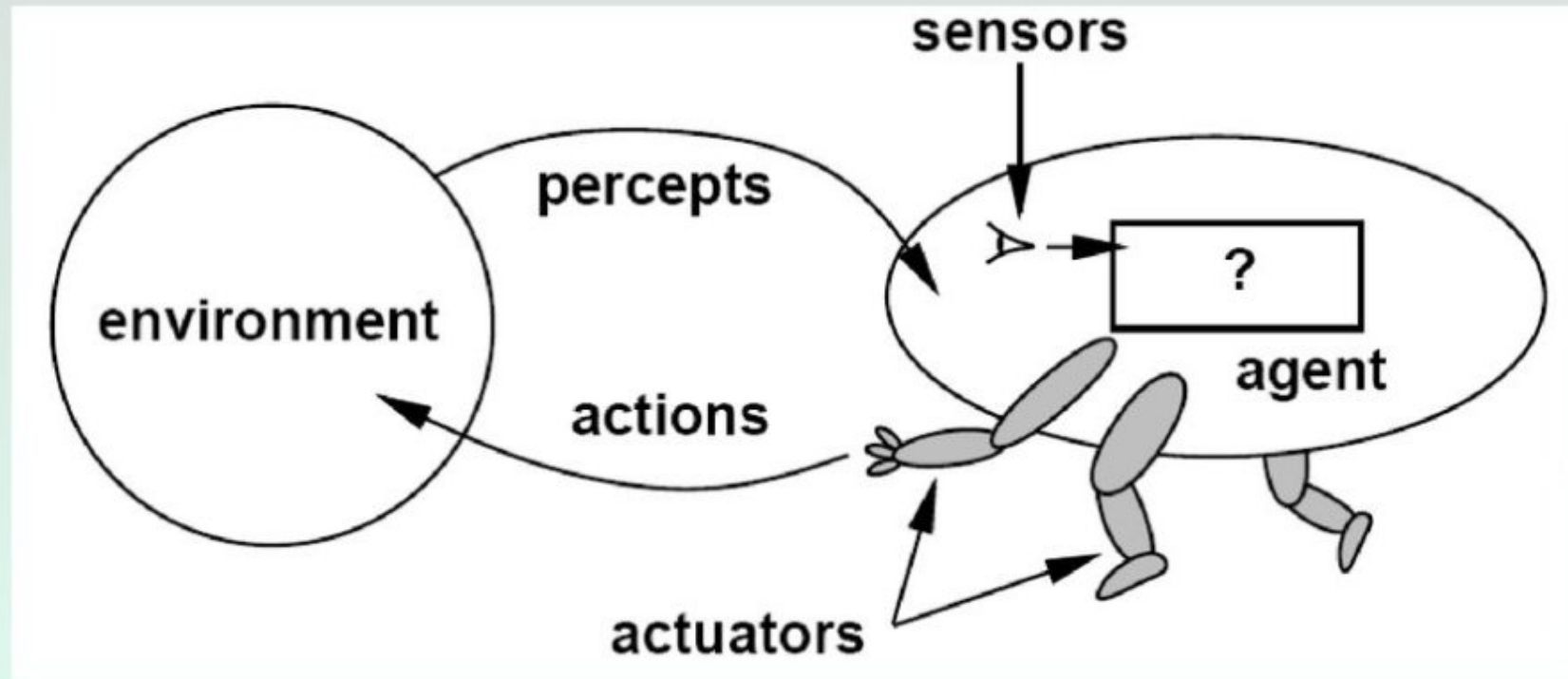


AI in Self-Driving Cars



Agents

An **agent** is anything that can be viewed as **perceiving** (getting input) its **environment** through **sensors** and **acting** upon that environment through **actuators** (tools used for actions).



- <https://pyimagesearch.com/2023/08/14/amazon-product-recommendation-systems/>
- https://www.amazon.in/BOBO-BIRD-Personalized-Wristwatches-Customized/dp/B07MBX4MTY/ref=sr_1_1?crid=KZNHECGP6MJ&dib=eyJ2IjojMSJ9.V7jrHYvYL2030tFshdu9yMEvXFAsfZjSc5d8rLFdBna8WNP2_l6GEkHAfqGJ5A3FDS-KlrEj_4zkd6A8DiNFISZrF_gnrjtfbvvMcL22eJGCYVfxPCsDcQTh7XZetZq57jIX06V4nL5rwQgaQ6Qgppt9kVEf3DyOumgit3s0gwpHjhkYFtjwQffs9_NACrMrPVLIXev84MwypNvIX-UMbbraPJ-Gds-_YZpkUZqGVzs5Y1aJfBTo3GKrsUybduxjFn60IDBzEz_wT6itIsvvc7mZcp28lI3n9627k5k5rsN46vicuvfnKxdrH919W83ALtaYxoDzCy2GH5Fg5jd-TGZev6tQu65GaGfwvXgBKyzSNNeEDmJl-wUO3F9Szm3k5MKifkwecb4RsGqA_UoJeujkpFrko_qH3zRXR8wTU2NebHXf9xmOT0gTAFkbKDN.zv98fM6xZ1e8IcRV0az0RpgqPihDRTITWNIJGtZdvg&dib_tag=se&keywords=watch%2Bfor%2Bman&qid=1735823759&sprefix=watch%2Caps%2C242&sr=8-1&th=1
- <https://www.vectorstock.com/royalty-free-vector/smart-house-icon-design-vector-7965710>

- https://www.google.com/search?q=face+recognition+lock+unlock+in+mobile+phone&sc_a_esv=dc598d9fff289246&rlz=1C1CHBD_enIN917IN917&udm=2&biw=1280&bih=585&ei=5Oh2Z4jhDMuY4-EP3PX0uAQ&ved=0ahUKEwil45DZ4teKAxVLzDgGHdw6HUcQ4dUDCBE&uact=5&oq=face+recognition+lock+unlock+in+mobile+phone&gs_lp=EgNpbWciLGZhY2UgcmlVjb2duaXRpb24gbG9jayB1bmxvY2sgaW4gbW9iaWxIHBob25lSL5EULsBWLpBcAJ4AJABAJgB6QGgAfURqgEGMC4xMy4yuAEDyAEA-AEBmAIAoAIAmAMAIAYBkgcAoAejBQ&sclient=img#vhid=8uaZjq-PCELNpM&vssid=mosaic
- <https://www.freecodecamp.org/news/build-an-ai-for-two-player-turn-based-games/>
- <https://viterbischool.usc.edu/news/2022/11/ai4health-a-collab-for-the-future-of-health-care/>
- <https://www.intuz.com/blog/guide-to-building-an-ai-chatbot>
- <https://dda.ndus.edu/ddreview/are-self-driving-cars-safe/>